



Albert Einstein Cup 2026

Skill 08: Mobile Applications Development

Session 3: NeubrandenBook Fans (Implementation & Functionality)

1. Introduction

The NeubrandenBook Catalogue app has a loyal readership. To deepen the connection between readers and their books, the publisher wants to allow users to explore reading previews before purchasing and provide a creative outlet for fans to share artwork of the stories they love.

Your task is to develop a tablet-optimised app featuring a *Main Page*, a *Book Details Page* (including a reading-preview modal and a fan-art section), and the specific interactions detailed below.

The designer has provided you with mockups (Mockups/). As an app developer, your task is to develop the corresponding app, strictly following the design of the provided mockups. Functions not mentioned in the functional requirements do not need to be implemented.

You have **3:00 hours** to implement the app.

2. General Demands

1. The app targets tablet devices and must support *portrait orientation only*.
2. The app shall be titled "NeubrandenBook Fans".
3. All text within quotation marks must be used exactly as written.
4. All specified animations must be fully implemented; placeholder stubs are not accepted.
5. Load all book data from the provided data file unless otherwise specified.

3. Page Demands

3.1. Global Elements

3.1.1. Elements

1. Header present on all pages containing:
 1. App Logo
 2. Title "NeubrandenBook"
 3. Subtitle "Discover Your Next Adventure"
 4. Search Bar with an icon and the label "Search books, authors, genres..."

3.1.2. Functional requirements

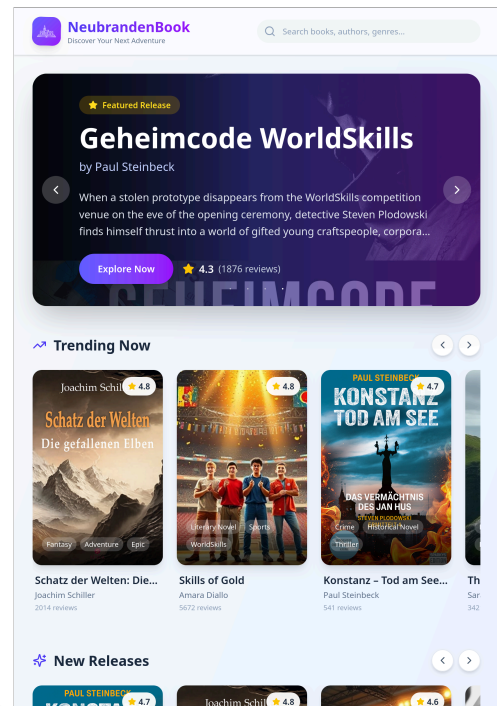
1. The logo must be animated as shown in the mockups.
2. Tapping the logo or the titles must navigate the user back to the *Main Page*.

The search bar is a visual element only and does not require functional implementation.

3.2. Main Page

3.2.1. Elements

1. Featured Banner Carousel containing:
 1. "Featured Release" chip with icon
 2. Book title and author
 3. Truncated book description
 4. Previous/Next control buttons
 5. "Explore Now" CTA button
 6. Star rating and review count
 7. Page indicator dots
2. Horizontal Book Lists "Trending Now", "New Releases", and "Editor's Picks", each with:
 1. Section icon
 2. Title
 3. Control buttons
 4. List of book items, each with:
 1. Thumbnail
 2. Title
 3. Genres
 4. Rating
 5. Author
 6. Review count



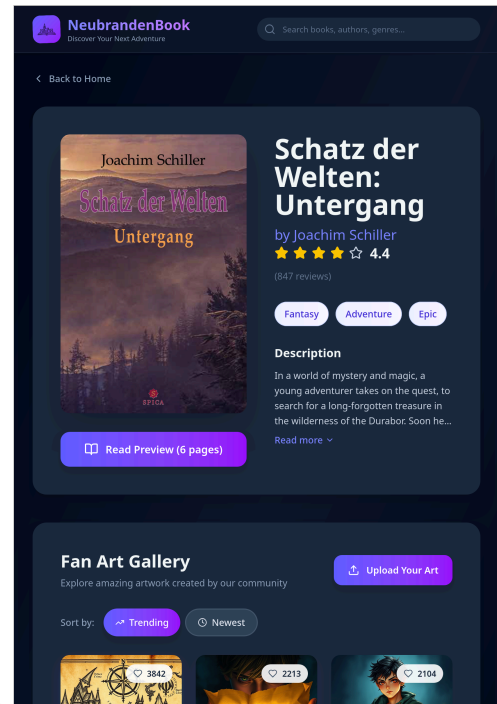
3.2.2. Functional requirements

1. Featured Banner Carousel:
 1. Auto-rotates every 5 seconds.
 2. Manual navigation (swiping or buttons) advances/retreats the carousel and resets the timer.
 3. Tapping "Explore Now" opens the *Book Details Page* for that book.
 4. Content transitions and the "Explore Now" interaction must be animated.
2. "Trending Now" list:
 1. Displays the top 20 books ordered by rating (descending).
 2. Includes all books regardless of sales volume.
3. "New Releases" list:
 1. Displays the 20 most recent books, ordered by publication date (descending).
4. "Editor's Picks" list:
 1. Displays a randomized selection of books flagged as editor_pick.
5. Horizontal Lists:
 1. Must support horizontal scrolling via swipe and control buttons.
 2. Tapping any book opens the *Book Details Page* via an animated transition.
6. Initial Load: The banner and list items must have an entry animation when the page opens.

3.3. Book Details Page

3.3.1. Elements

1. Book Details Section:
 1. Book cover image
 2. Book title and author name
 3. Star rating display and average rating value
 4. Genre chips
 5. Book description (collapsed to 4 lines by default)
 6. "Read more" / "Show less" toggle
 7. "Read Preview (X pages)" button
2. Fan Art Section:
 1. Title: "Fan Art Gallery"
 2. Subtitle: "Explore amazing artwork created by our community"
 3. Sort controls: "Trending" / "Newest" chips with icons
 4. "Upload Your Art" button
 5. A 3-column masonry grid of images. Each item includes:
 1. The artwork image
 2. Like and share counters
 3. Artist name



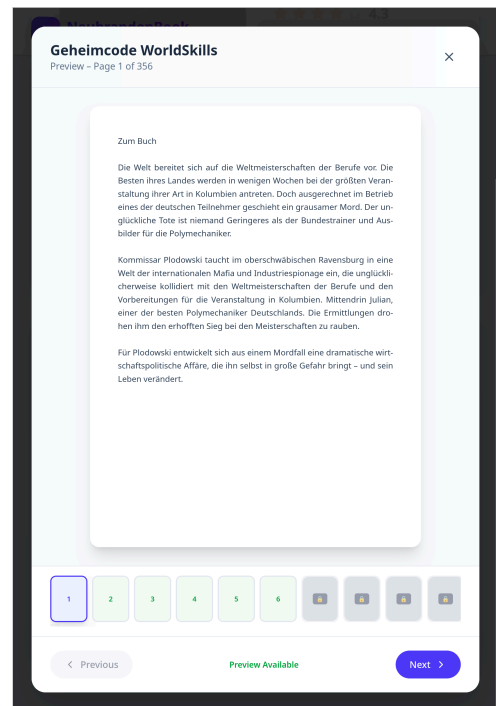
3.3.2. Functional requirements

1. Book Description: Tapping "Read more" expands the text fully and changes the label to "Show less"; tapping "Show less" collapses it back to 4 lines.
2. Read Preview: Tapping the button opens the *Reading Preview Modal* with an animation.
3. Fan Art Section:
 1. Sort "Trending": Ordered by likes (descending), then by upload date (descending).
 2. Sort "Newest": Ordered by upload date (descending).
 3. Tapping an image opens the *Fan Art Fullscreen Viewer* with an animation.
 4. Long-pressing an image opens a context menu with "Share" and "Save to Gallery":
 1. "Share" opens the system share sheet.
 2. "Save to Gallery" saves the image to the device library and shows a "Saved to Gallery" toast.
 5. The "Upload Your Art" button opens the *Fan Art Upload Flow*.
 6. Like/Share counters must update in real-time within the current session.
4. *Initial Load*: The two main sections (Details and Fan Art) must have entry animations.

3.4. Reading Preview Modal

3.4.1. Elements

1. Modal Overlay appearing above the *Book Details Page*
2. Book Title
3. Page Indicator: "Preview – Page X of Y"
4. Close Button
5. Content Area:
 1. Readable, full-resolution preview pages
 2. OR a placeholder if unavailable:
 1. Title "Preview Not Available"
 2. Subtitle "Purchase the full book to continue reading"
6. Bottom Page Slider: Overview of all pages (locked pages must be blurred with a lock icon overlay)
7. "Previous" and "Next" Buttons (disabled on the first and last pages, respectively)
8. Status Indicator: "Preview Available" or "Purchase Required"



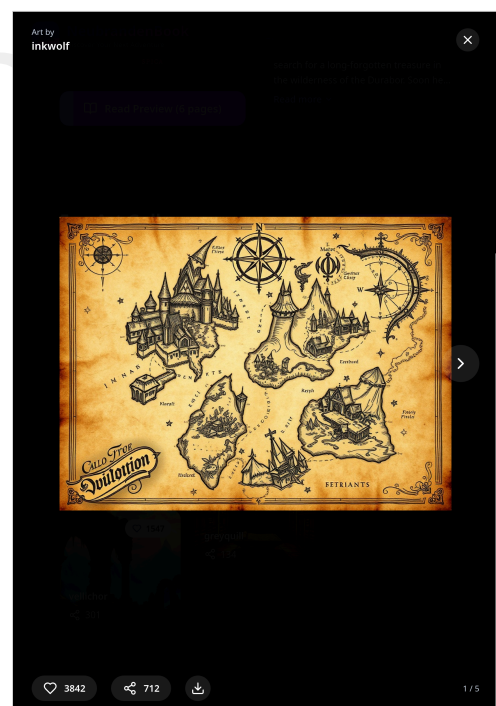
3.4.2. Functional requirements

1. The modal must open with a fade-scale animation and reverse it upon closing.
2. Preview pages are interactive; all other pages are blurred and non-interactive.
3. Total page count must be retrieved from the book data file.
4. Horizontal swiping allows navigation between pages. Swiping past the last preview page must trigger a "bounce" animation.
5. Tapping a page in the bottom slider selects that page via an animation.

3.5. Fan Art Fullscreen Viewer

3.5.1. Elements

1. Full-screen image display
2. Artist credit: "Art by [Artist Name]"
3. Like button with counter
4. Share button with counter
5. Save button
6. Page indicator (e.g., "3 / 12")
7. Close button and navigation controls



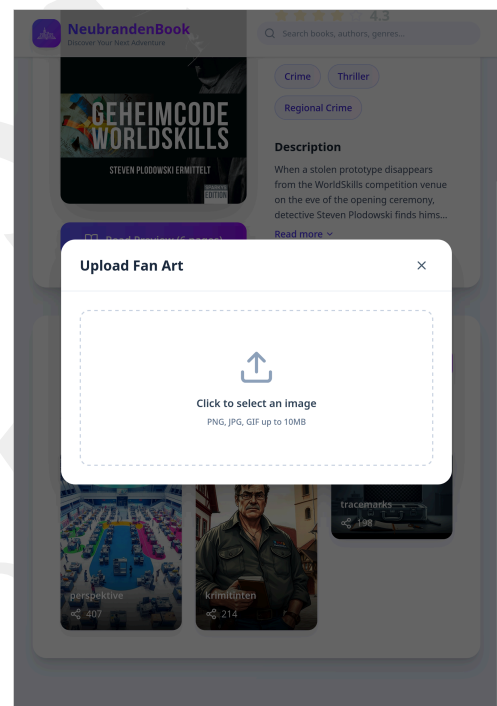
3.5.2. Functional requirements

1. The viewer must open with a fade transition from the tapped thumbnail; closing reverses this.
2. Navigation: Users can swipe horizontally or use control buttons to browse all fan art for the book (maintaining the current sort order).
3. The page indicator must update dynamically during navigation.
4. Interaction:
 1. Pinch-to-zoom is required.
 2. Double-tapping the image triggers a (un)like action with an animated heart at the tap position.
 3. The Like button toggles state with an animation.
 4. The Share button triggers the system share sheet; the counter increments upon confirmation.
 5. The Save button saves the image and shows a “Saved to gallery” toast.

3.6. Fan Art Upload Flow

3.6.1. Elements

1. Title: “Upload Fan Art”
2. Close button
3. Image picker area
4. Preview of the selected image
5. “Upload to Gallery” confirmation button
6. “Cancel” button



3.6.2. Functional requirements

1. The modal must open with a fade-and-scale transition; closing reverses this.
2. Tapping the picker opens the device gallery for single image selection.
3. Selected images must be shown as a preview before confirmation.
4. Tapping “Upload to Gallery” adds the image to the book’s list:
 1. In “Newest” sort: Image is inserted at the top.
 2. In “Trending” sort: Image is appended to the end.
5. The image should have your name as the artist and start with zero likes and shares.
6. Tapping “Cancel” or the “X” button on the image clears the selection and returns the user to the *Book Details Page*.

4. Submission

Your submission should contain the following top-level files and directories:

- the compiled APK with package name `de.blzit.aec2026.skill08.session3.competitorX`
- a directory with the full source code
- optional: a README file containing notes for the marking

These files and directories should be pushed to the Git repository `Session-3_Competitor8-X`. Only the latest contents of the `main` branch will be taken into account.

